Activation functions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable activation and chain trace.

Compare fixed functions,explain nonlinear information loss and state unsupported claims.

Keep one habit: W1–W3/W5 → selected new chain → project activation record → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Representing **input**s as vectors. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- First compute an affine score z = w dot x + b. An activation then returns a = f(z). In a network this value may become an **input** to another layer; it is not automatically a class **prediction**.

  We will compare three functions using the same incoming scores so that only the activation changes.

- Identity returns z unchanged. **ReLU**, pronounced ree-loo, returns the larger of zero and z. Thus **ReLU**(-2) = 0, **ReLU**(0) = 0 and **ReLU**(2) = 2. Positive scores keep their magnitude; **ReLU** does not turn every positive value into 1.

- These are fixed arithmetic functions. Selecting a function or changing its **input** is not itself a **training** procedure. Later weeks will introduce losses and parameter updates.

Bias terms and decision boundaries

> **My remembered idea, example and correction**

Representing inputs as vectors

> **My remembered idea, example and correction**

Lesson reference for workbook practice

Incoming z	Identity output	ReLU output
-2	-2	0
0	0	0
2	2	2

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Same incoming z−2/0/2:identity−2/0/2, ReLU0/0/2, sigmoid≈.1192/.5/.8808. Declared class 1if sigmoid≥.5 gives0/1/1; strict>changes equality only. ReLU is nonlinear:outputs at−2/2 average1 but output at midpoint 0is0. Identity chain h2x+1, y3h−2 collapses to6x+1; ReLU instead can change the function family. Separate pictured chain z=x−3, hReLU(z), y3h+2:inputs 1/3/4 give z−2/0/1, h0/0/1, y2/2/5; identity gives y−4/2/5.

  Functions are supplied, not trained; bounded sigmoid output alone is not calibration evidence. Independent chain uses its own different constants.

- For scores -2, 0, 2, list identity and **ReLU output**s. Explain why **ReLU**(2) is not 1.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Same incoming z−2/0/2:identity−2/0/2, ReLU0/0/2, sigmoid≈.1192/.5/.8808. Declared class 1if sigmoid≥.5 gives0/1/1; strict>changes equality only. ReLU is nonlinear:outputs at−2/2 average1 but output at midpoint 0is0. Identity chain h2x+1, y3h−2 collapses to6x+1; ReLU instead can change the function family. Separate pictured chain z=x−3, hReLU(z), y3h+2:inputs 1/3/4 give z−2/0/1, h0/0/1, y2/2/5; identity gives y−4/2/5.

  Functions are supplied, not trained; bounded sigmoid output alone is not calibration evidence. Independent chain uses its own different constants.

- Use the lesson **sigmoid** values for -2, 0, 2. Predict classes under a >= 0.5, then explain the zero-score case.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Same incoming z−2/0/2:identity−2/0/2, ReLU0/0/2, sigmoid≈.1192/.5/.8808. Declared class 1if sigmoid≥.5 gives0/1/1; strict>changes equality only. ReLU is nonlinear:outputs at−2/2 average1 but output at midpoint 0is0. Identity chain h2x+1, y3h−2 collapses to6x+1; ReLU instead can change the function family. Separate pictured chain z=x−3, hReLU(z), y3h+2:inputs 1/3/4 give z−2/0/1, h0/0/1, y2/2/5; identity gives y−4/2/5.

  Functions are supplied, not trained; bounded sigmoid output alone is not calibration evidence. Independent chain uses its own different constants.

- Compare **ReLU** at **input**s -2, 0, 2. Give the average of the two endpoint **output**s and the **output** at the midpoint; explain what this shows.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Same incoming z−2/0/2:identity−2/0/2, ReLU0/0/2, sigmoid≈.1192/.5/.8808. Declared class 1if sigmoid≥.5 gives0/1/1; strict>changes equality only. ReLU is nonlinear:outputs at−2/2 average1 but output at midpoint 0is0. Identity chain h2x+1, y3h−2 collapses to6x+1; ReLU instead can change the function family. Separate pictured chain z=x−3, hReLU(z), y3h+2:inputs 1/3/4 give z−2/0/1, h0/0/1, y2/2/5; identity gives y−4/2/5.

  Functions are supplied, not trained; bounded sigmoid output alone is not calibration evidence. Independent chain uses its own different constants.

- A model returns a **sigmoid** value 0.9 on one case. Does the numerical range alone establish a calibrated 90 percent probability? Explain what needs checking.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Let z = x - 2, a = **ReLU**(z), y = 2a + 1. For x = 0, 2, 5, record every z, a and y. Then replace **ReLU** with identity and give all three y values. Which **input**s have the same **ReLU output**, and why?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Trace a new chain independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled **activation function**s **evidence** sheet to Image-pattern Network Investigation. Show how you can trace identity **relu** and **sigmoid** on a numerical score.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Apply a declared class rule for **Activation function**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Activation function**s. Proposed explanation: "**ReLU** returns a class label". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Let z = x - 2, a = **ReLU**(z), y = 2a + 1. For x = 0, 2, 5, record every z, a and y. Then replace **ReLU** with identity and give all three y values. Which **input**s have the same **ReLU output**, and why?

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

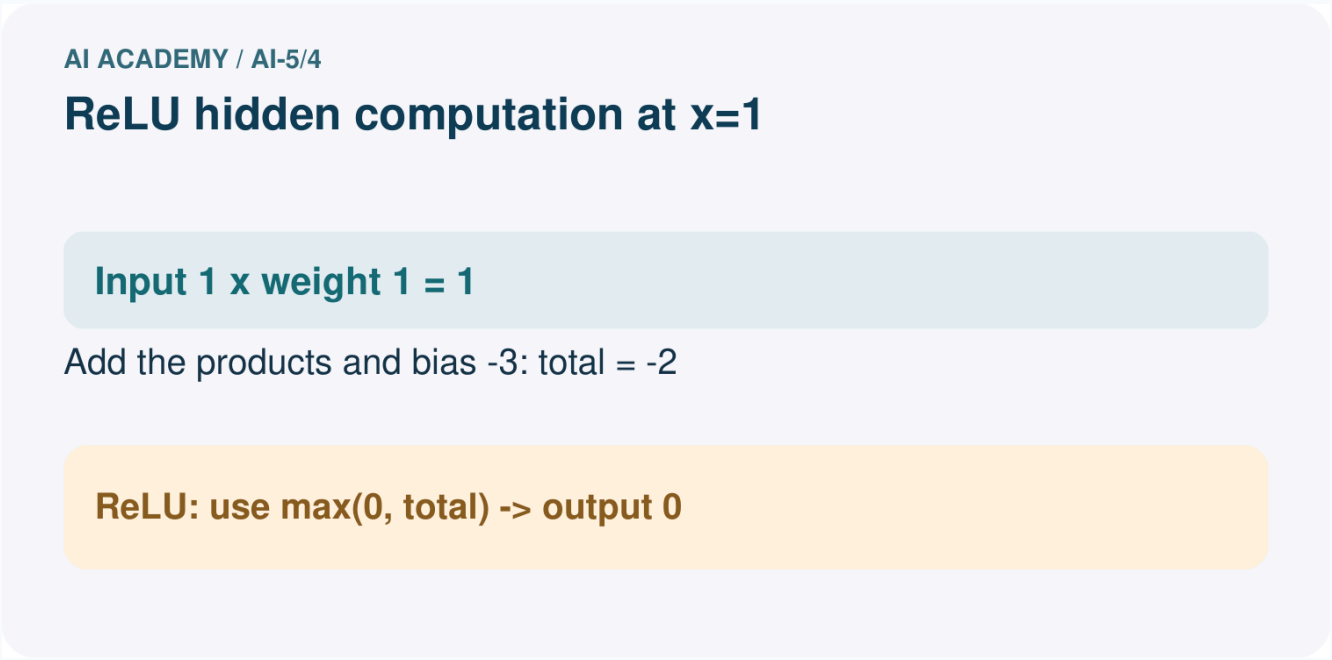
- See what **ReLU** removes before the next weighted step.
- Toy network z=x-3; h=max(0, z); y=3h+2. **Input**s x=1, 3, 4.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- Which original **input**s cannot be distinguished by this network's final **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/4

Activation functions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute preactivation

z values are -2, 0, 1.



2. Apply ReLU

h values are 0, 0, 1; the negative value is replaced by0.



3. Compute output

y values are 2, 2, 5; x = 1 and x = 3 collapse to the same output.

CHANGE ONE THING

Replace ReLU with identity.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- See what **ReLU** removes before the next weighted step.
- Which labelled step matches this detail? y values are 2, 2, 5; x = 1 and x = 3 collapse to the same **output**.
- 1. Apply **ReLU**
- 2. Compute **output**
- 3. Compute preactivation
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compute **output** y values are 2, 2, 5; x = 1 and x = 3 collapse to the same **output**. **ReLU** means keep a positive value and replace a negative one with zero.

The **output** layer receives that transformed value, not the original **input**. **Input**s one and three therefore both deliver zero to the final layer and produce two.

Replacing **ReLU** with identity changes the procedure: negative intermediate values now remain available to the next layer.

Changed-case question and optional hint

- Changed-case practice: Replace **ReLU** with identity. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. h becomes -2, 0, 1 and **output**s become -4, 2, 5.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: h becomes -2, 0, 1 and **output**s become -4, 2, 5. Identity retains the negative intermediate value instead of clipping it.

Repair and retain the record

Review Bias terms and decision boundaries (ai-5/3). Goal: Calculate a weighted score with a bias term. Short practice: Is B positive under the original or changed model?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Representing **input**s as vectors (ai-5/1). Goal: Encode a record using a fixed feature order and units. Short practice: Why does equal total brightness not make A and B identical **input**s?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Activation functions: the case

AI ACADEMY / AI-5/4 · 01 · LOOK AT THE FACTS

Activation functions: the case

The situation

See what ReLU removes before the next weighted step.

Supplied evidence

Toy network z=x-3; h=max(0, z); y=3h+2.
Inputs x=1, 3, 4.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain activation functions using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

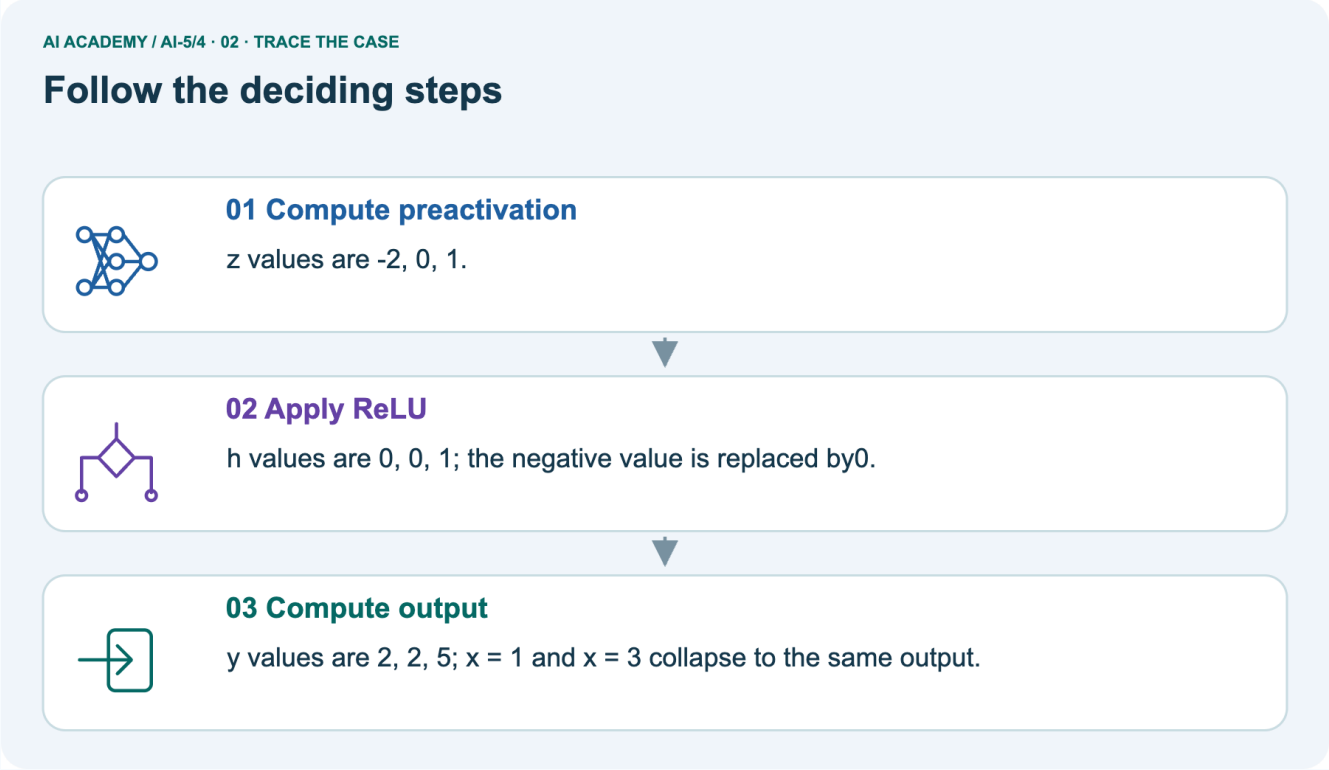
See what **ReLU** removes before the next weighted step. Supplied facts: Toy network z=x-3; h=max(0, z); y=3h+2. **Input**s x=1, 3, 4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



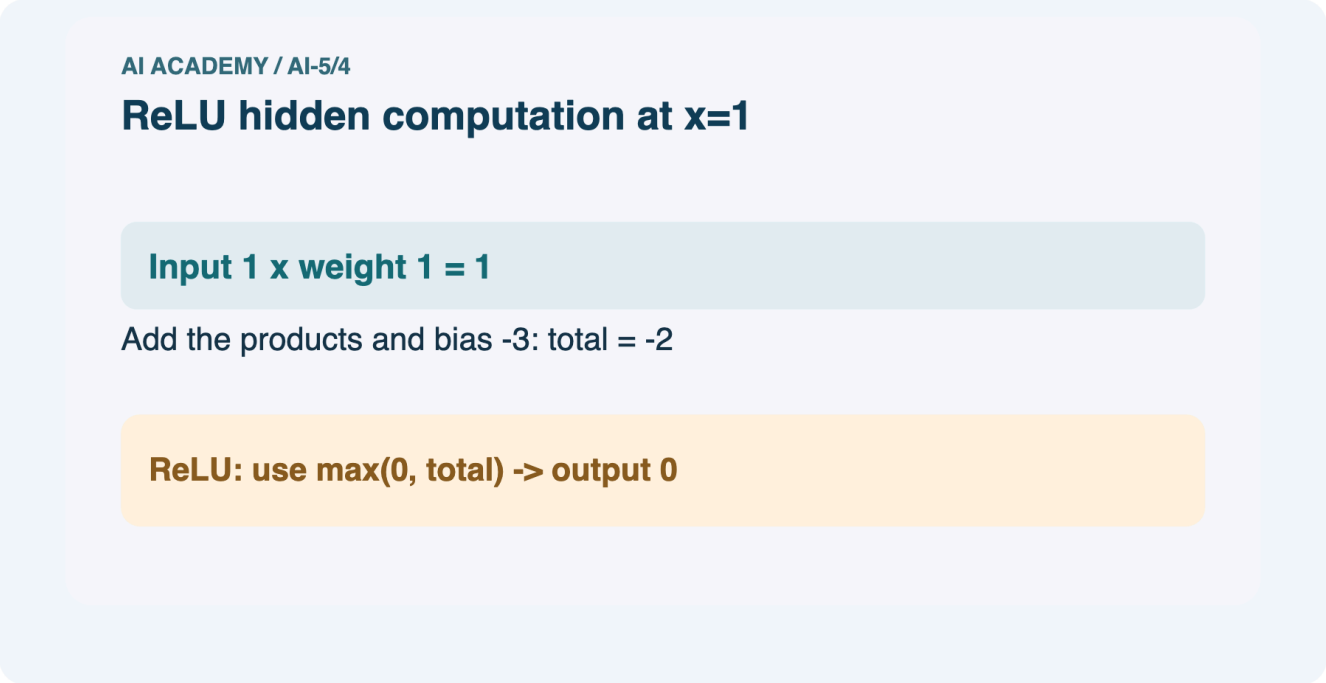
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute preactivation: z values are -2, 0, 1. → Apply **ReLU**: h values are 0, 0, 1; the negative value is replaced by0. → Compute **output**: y values are 2, 2, 5; x = 1 and x = 3 collapse to the same **output**.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. ReLU hidden computation at x=1



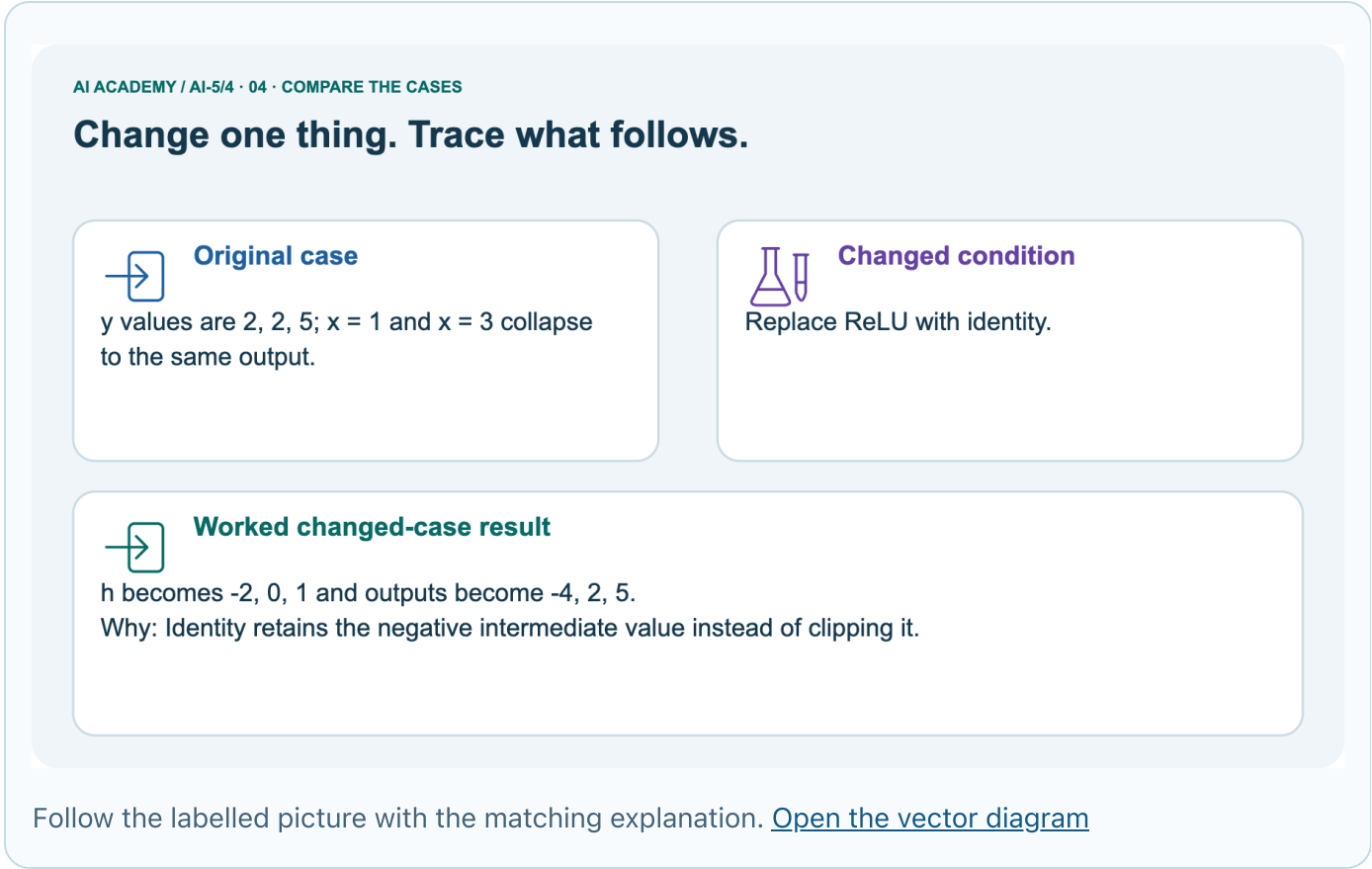
Follow the labelled picture with the matching explanation. Open the vector diagram

ReLU hidden computation at x=1. **Input**s: [1]; weights: [1]; bias: -3; activation: **relu**; **output**: 0.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

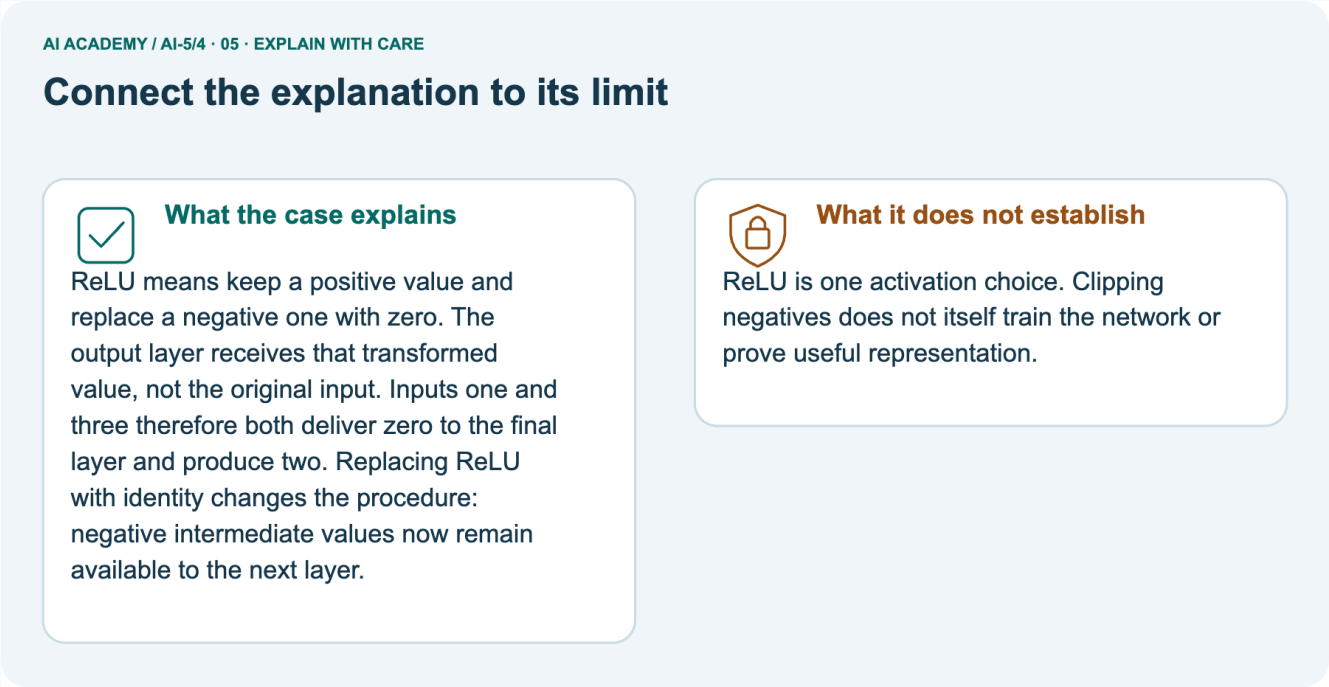
Original case: y values are 2, 2, 5; x = 1 and x = 3 collapse to the same **output**. Changed condition: Replace **ReLU** with identity. Expected result: h becomes -2, 0, 1 and **output**s become -4, 2, 5.

Reason: Identity retains the negative intermediate value instead of clipping it.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

ReLU means keep a positive value and replace a negative one with zero. The **output** layer receives that transformed value, not the original **input**.

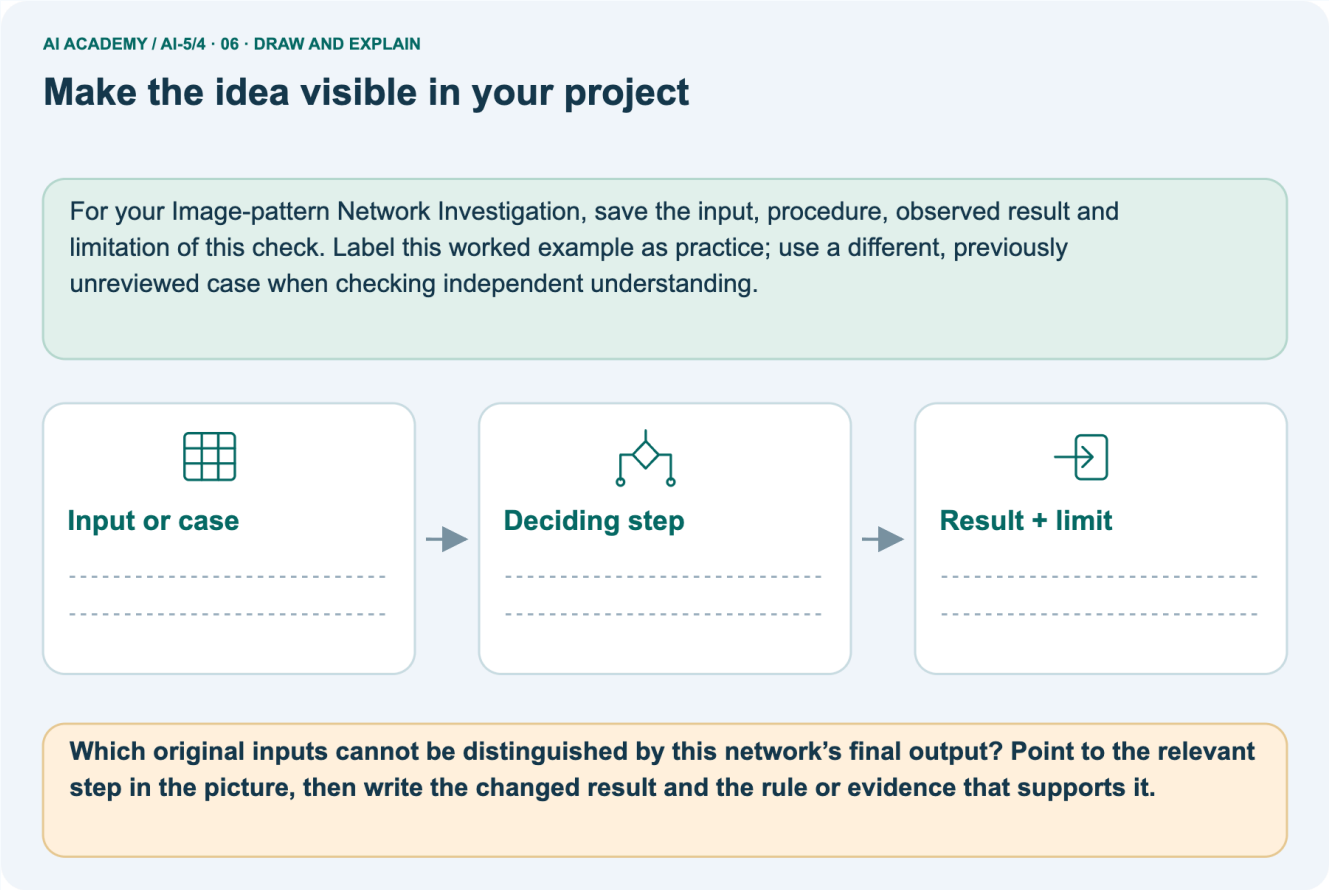
Inputs one and three therefore both deliver zero to the final layer and produce two. Replacing **ReLU** with identity changes the procedure: negative intermediate values now remain available to the next layer.

Limit: **ReLU** is one activation choice. Clipping negatives does not itself train the network or prove useful representation.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which original **input**s cannot be distinguished by this network's final **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Activation functions: the case

02. Follow the deciding steps

03. ReLU hidden computation at x=1

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.